

Reinforcement Learning

Chapter 2: Model-based RL

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Classical RL Methods: Recall

Ultimate goal in an RL problem is to find the *optimal policy*

As mentioned, we have two *major challenges* in this way

- 1 We need to compute *values* explicitly
- 2 We often deal with settings with *huge state spaces*?

In this part of the course, we are going to handle the first challenge

- This chapter $\rightsquigarrow$ Model-based methods
- Next chapter $\rightsquigarrow$ Model-free methods

A Good Start Point: *Model-based RL*

In **model-based** methods

we are able to mathematically describe the environment

This might come from the *nature* or simply *postulated by us*

Model-Based RL

Bellman Equation

value iteration

policy iteration

Model-free RL

on-policy methods

temporal difference

Monte Carlo

SARSA

off-policy methods

Q-learning

Describing Markov Trajectory

Markovity of the state indicates that we observe the following trajectory

$$S_0, A_0 \rightarrow (R_1, S_1), A_1 \rightarrow \dots \rightarrow (R_t, S_t), A_t \rightarrow (R_{t+1}, S_{t+1})$$

This trajectory describes a **Markov process** with conditional distribution

$$\begin{aligned} p(r, \bar{s} | s, a) &= \Pr \{ R_{t+1} = r, S_{t+1} = \bar{s} | S_t = s, A_t = a \} \\ &= \Pr \{ R_t = r, S_t = \bar{s} | S_{t-1} = s, A_{t-1} = a \} \\ &\vdots \\ &= \Pr \{ R_1 = r, S_1 = \bar{s} | S_0 = s, A_0 = a \} \end{aligned}$$

*The above trajectory describes a **Markov Decision Process (MDP)***

Finite MDPs

Finite MDP

The Markov process

$$S_0, A_0 \rightarrow (R_1, S_1), A_1 \rightarrow \dots \rightarrow (R_t, S_t), A_t$$

is a finite MDP if *rewards*, *actions* and *states* belong to a finite set, i.e.,

$$r \in \{r^1, \dots, r^L\} \quad a \in \{a^1, \dots, a^M\} \quad s \in \{s^1, \dots, s^N\}$$

MDPs are completely described by conditional distribution $p(r, \bar{s} | s, a)$

We call $p(r, \bar{s} | s, a)$ hereafter *rewarding-transition model*

Model-based RL via MDP

Rewarding Model

In state $S_t = s$, we act $A_t = a$; then, distribution R_{t+1} is

$$p(r|s, a) = \sum_{n=1}^N p(r, s^n|s, a)$$

Transition Model

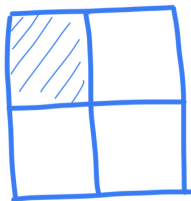
In state $S_t = s$, we act $A_t = a$; then, distribution of next state S_{t+1} is

$$p(\bar{s}|s, a) = \sum_{\ell=1}^L p(r^\ell, \bar{s}|s, a)$$

Example: Dummy Grid World

We have a grid board where at each cell we can move

$$\mathcal{A} = \{0 \equiv \text{left}, 1 \equiv \text{down}, 2 \equiv \text{right}, 3 \equiv \text{up}\}$$



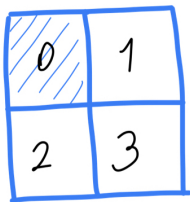
Our ultimate goal is to arrive at **top-left corner**

through **shortest** path

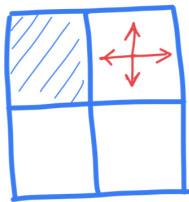
This problem describes an MDP with **deterministic rewarding-transition model**

- **State** is the cell index
- **Action** is the direction we move
- **Reward** is -1 if we move and -0.5 if we **hit a corner**

Example: *Dummy Grid World*



$$\mathcal{S} = \{0, 1, 2, 3\}$$

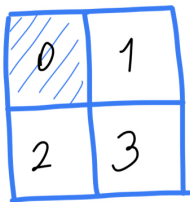


$$\mathcal{A} = \{0 \equiv \text{left}, 1 \equiv \text{down}, 2 \equiv \text{right}, 3 \equiv \text{up}\}$$

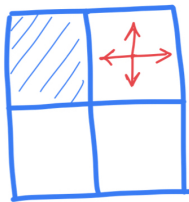
Let's write the *rewarding-transition model* down

$$p(r, \bar{s} | \mathbf{3}, \mathbf{3}) = \begin{cases} 1 & (r, \bar{s}) = (-1, 1) \\ 0 & (r, \bar{s}) \neq (-1, 1) \end{cases}$$

Example: *Dummy Grid World*



$$\mathcal{S} = \{0, 1, 2, 3\}$$



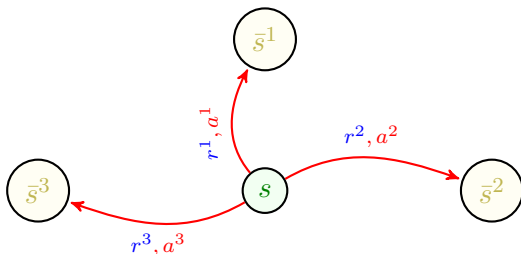
$$\mathcal{A} = \{0 \equiv \text{left}, 1 \equiv \text{down}, 2 \equiv \text{right}, 3 \equiv \text{up}\}$$

Let's write the *rewarding-transition model* down

$$p(r, \bar{s} | 0, a) = \begin{cases} 1 & (r, \bar{s}) = (0, 0) \\ 0 & (r, \bar{s}) \neq (0, 0) \end{cases} \rightsquigarrow s = 0 \text{ is terminal state}$$

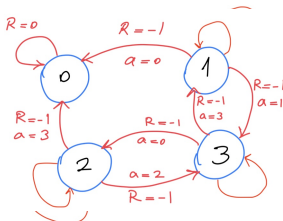
Transition Diagram

It is sometimes helpful to show **transitions** via a *transition diagram*



- Each node is a **possible state**: we have in total N nodes
- Node s is connected to $\bar{s}$ if the probability of transition is **non-zero**
 - ↳ We could specify the **action** that **can** lead us to the **new state**
 - ↳ The graph could have **loops** including **self-loop**

Transition Diagram: *Dummy Grid World*



In our dummy grid world, we have *four states*

- Once in *terminal state*, we always remain there with *no rewards*
- From *state* $s = 1$ we can go to *states* $\bar{s} = 0, 3$ or hit a corner
- From *state* $s = 2$ we can go to *states* $\bar{s} = 0, 3$ or hit a corner
- From *state* $s = 3$ we can go to *states* $\bar{s} = 1, 2$ or hit a corner

Expected Action Reward

Rewarding-transition model describes the environment completely

$$\begin{aligned}\bar{\mathcal{R}}(s, a) &= \mathbb{E} \{R_{t+1} | s, a\} \rightsquigarrow \text{we simplify notation } S_t = s \text{ to } s \\ &= \sum_{\ell=1}^L r^{\ell} p(r^{\ell} | s, a) \\ &= \sum_{\ell=1}^L r^{\ell} \sum_{n=1}^N p(r^{\ell}, s^n | s, a) \\ &= \sum_{\ell=1}^L \sum_{n=1}^N r^{\ell} p(r^{\ell}, s^n | s, a) \rightsquigarrow \text{rewarding-transition model}\end{aligned}$$

Expected Action Reward

Expected Action Reward

The expected reward for a *state-action pair* (s, a) is defined as

$$\bar{\mathcal{R}}(s, a) = \mathbb{E} \{ R_{t+1} | s, a \} = \sum_{\ell=1}^L \sum_{n=1}^N r^{\ell} p(r^{\ell}, s^n | s, a)$$

Obviously, $\bar{\mathcal{R}}(s, a)$ does *not* depend on *policy*

Expected Policy Reward

- + Can we relate it also to *our policy*?
- Sure! We could *average over our policy*

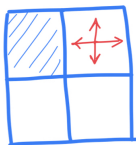
Expected Policy Reward

The expected immediate reward of policy π at *state* s is defined as

$$\begin{aligned}\bar{\mathcal{R}}_{\pi}(s) &= \mathbb{E}_{\pi} \{R_{t+1} | s\} = \sum_{m=1}^M \mathbb{E} \{R_{t+1} | s, a^m\} \pi(a^m | s) \\ &= \sum_{m=1}^M \sum_{\ell=1}^L \sum_{n=1}^N r^{\ell} p(r^{\ell}, s^n | s, a) \pi(a^m | s)\end{aligned}$$

It describes reward we expect to see immediately after *state* s while playing π

Example: Dummy Grid World



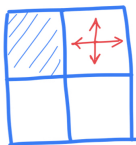
In our dummy grid world, we can easily compute the *expected immediate reward*

$$\bar{\mathcal{R}}(1, a) = \begin{cases} -1 & a \in \{0, 1\} \\ -0.5 & a \in \{2, 3\} \end{cases}$$

Obviously in *terminal state* we always get *zero expected reward*, e.g., for *all* a

$$\bar{\mathcal{R}}(0, a) = 0$$

Example: Dummy Grid World



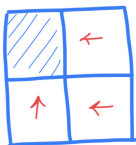
Now assume that we play *uniformly at random*, i.e., for all a and s

$$\pi(a|s) = \frac{1}{4}$$

In this case the *expected policy reward* is

$$\bar{\mathcal{R}}_{\pi}(1) = \sum_{a=0}^3 \bar{\mathcal{R}}(1, a) \pi(a|1) = -0.75$$

Example: Dummy Grid World



But if we change to *above deterministic* policy: the *expected reward* changes to

$$\bar{\mathcal{R}}_{\pi}(1) = \sum_{a=0}^3 \bar{\mathcal{R}}(1, a) \pi(a|1) = \bar{\mathcal{R}}(1, 0) = -1$$

and we can easily show that

$$\bar{\mathcal{R}}_{\pi}(0) = 0 \quad \bar{\mathcal{R}}_{\pi}(2) = -1 \quad \bar{\mathcal{R}}_{\pi}(3) = -1$$

Computing Value Functions: *Naive Approach*

What about the value function?!

$$\begin{aligned}v_{\pi}(s) &= \mathbb{E}_{\pi} \{G_t | s\} \\&= \mathbb{E}_{\pi} \{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | s\} \\&= \mathbb{E}_{\pi} \{R_{t+1} | s\} + \gamma \mathbb{E}_{\pi} \{R_{t+2} | s\} + \gamma^2 \mathbb{E}_{\pi} \{R_{t+3} | s\} + \dots \\&= \bar{\mathcal{R}}_{\pi}(s) + \gamma \mathbb{E}_{\pi} \{R_{t+2} | s\} + \gamma^2 \mathbb{E}_{\pi} \{R_{t+3} | s\} + \dots\end{aligned}$$

- + *How can we compute next terms?*
- We could use the *rewarding-transition model of MDP*

Computing Value Functions: *Naive Approach*

Let's try the second term for example: we first define the notation

$$\mathbb{E}_{\pi} \{R_{t+2} | s, s^n, a^m, a^j\} = \mathbb{E}_{\pi} \{R_{t+2} | S_t = s, S_{t+1} = s^n, A_t = a^m, A_{t+1} = a^j\}$$

We can easily compute $\mathbb{E}_{\pi} \{R_{t+2} | s, s^n, a^m, a^j\}$ as

$$\begin{aligned} \mathbb{E}_{\pi} \{R_{t+2} | s, s^n, a^m, a^j\} &= \sum_{\ell=1}^L r^{\ell} p \left(r^{\ell} | s, s^n, a^m, a^j \right) \\ &= \sum_{\ell=1}^L r^{\ell} p \left(r^{\ell} | s^n, a^j \right) \end{aligned}$$

Computing Value Functions: Naive Approach

We can then say that

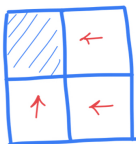
$$\mathbb{E}_{\pi} \{R_{t+2}|s\} = \sum_{n=1}^N \sum_{m=1}^M \sum_{j=1}^M \mathbb{E}_{\pi} \{R_{t+2}|s, s^n, a^m, a^j\} p(a^m, s^n, a^j|s)$$

and write down $p(a^m, s^n, a^j|s)$ using chain rule

$$\begin{aligned} p(a^m, s^n, a^j|s) &= p(a^m|s) p(s^n|s, a^m) p(a^j|s, a^m, s^n) \\ &= \pi(a^m|s) \underbrace{p(s^n|s, a^m)}_{\text{transition model}} \pi(a^j|s^n) \end{aligned}$$

- + How can we compute the next term?
- We should repeat the same approach: there will be *more nested sums*

Example: *Dummy Grid World*



Let's start with the *above policy*: π^1

$$v_{\pi^1}(1) = \mathbb{E}_{\pi^1} \{ R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | 1 \}$$

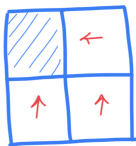
Following policy at $s = 1$ we end up at *terminal state* at next time

$$v_{\pi^1}(1) = \mathbb{E}_{\pi^1} \{ R_{t+1} + \gamma 0 + \gamma^2 0 + \dots | 1 \} = \bar{\mathcal{R}}_{\pi^1}(1) = -1$$

Same way, we can conclude that

$$v_{\pi^1}(0) = 0 \quad v_{\pi^1}(2) = -1 \quad v_{\pi^1}(3) = -2$$

Example: *Dummy Grid World*

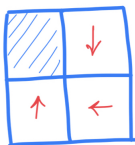


Let's change to *policy* π^2 : we could follow same steps to show that

$$v_{\pi^2}(0) = 0 \quad v_{\pi^2}(1) = -1 \quad v_{\pi^2}(2) = -1 \quad v_{\pi^2}(3) = -2$$

We note that it returns the same values as *policy* π^1

Example: *Dummy Grid World*



Let's now look at *policy* π^3 : we could follow same steps to show that

$$v_{\pi^3}(0) = 0 \quad v_{\pi^3}(1) = -3 \quad v_{\pi^3}(2) = -1 \quad v_{\pi^3}(3) = -2$$

We can see that

$$\pi^1 = \pi^2 \geq \pi^3$$

Computing Value Functions: *Practical Approach*

- + *But, we should compute **infinite** terms **in general**!*
- *Well, if we are lucky: the sequence either **terminates** or **shows a pattern***
- + *What if that doesn't happen?*
- *Then, this approach really does **not** work!*

*This is why we called it the **naive approach**, since we **never** use this approach: in practice, we always invoke*

Bellman equation

*and find the value via **dynamic programming***

Future Return: *Recursive Property*

Even though **future return** looks **infinte**, it has a simple recursive property

$$\begin{aligned} G_t &= \sum_{i=0}^{\infty} \gamma^i R_{t+i+1} \\ &= R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \\ &= R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \dots) \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

We can use this property to find a **fixed-point equation** for the value function!

Value Function: Recursive Property

Say we are playing with policy π : we can write the value function as

$$\begin{aligned}
 v_{\pi}(s) &= \mathbb{E}_{\pi} \{G_t | s\} \\
 &= \mathbb{E}_{\pi} \{R_{t+1} + \gamma G_{t+1} | s\} \\
 &= \mathbb{E}_{\pi} \{R_{t+1} | s\} + \gamma \mathbb{E}_{\pi} \{G_{t+1} | s\} \\
 &= \bar{\mathcal{R}}_{\pi}(s) + \gamma \underbrace{\mathbb{E}_{\pi} \{G_{t+1} | s\}}_{?}
 \end{aligned}$$

- + Isn't that term again the value function at s ?
- Be careful! It's **not**

Attention

The second term is not the value of **state** s

$$\mathbb{E}_{\pi} \{G_{t+1} | s\} = \mathbb{E}_{\pi} \{G_{t+1} | S_t = s\} \neq \mathbb{E}_{\pi} \{G_t | S_t = s\} = v_{\pi}(s)$$

Value Function: *Recursive Property*

Let's do some marginalization

$$\begin{aligned}\mathbb{E}_{\pi} \{G_{t+1}|s\} &= \sum_{n=1}^N \mathbb{E}_{\pi} \{G_{t+1}|S_t = s, S_{t+1} = s^n\} \Pr \{S_{t+1} = s^n|S_t = s\} \\ &= \sum_{n=1}^N \mathbb{E}_{\pi} \{G_{t+1}|s, s^n\} p(s^n|s)\end{aligned}$$

Well, we need to specify the two terms in under summation, i.e.,

- $\mathbb{E}_{\pi} \{G_{t+1}|s, s^n\}$
- $p(s^n|s) = \Pr \{S_{t+1} = s^n|S_t = s\}$

Value Function: Recursive Property

Recall the trajectory

$$\dots \rightarrow (R_t, S_t = s), A_t \rightarrow (R_{t+1}, S_{t+1} = s^n), A_{t+1} \rightarrow (R_{t+2}, S_{t+2})$$

If we know *state* S_{t+1} any reward after $t + 1$ *only* depends on S_{t+1} , i.e.,

$$\mathbb{E}_{\pi} \{G_{t+1} | S_t = s, S_{t+1} = s^n\} = \mathbb{E}_{\pi} \{G_{t+1} | S_{t+1} = s^n\}$$

This indicates that

$$\mathbb{E}_{\pi} \{G_{t+1} | s, s^n\} = v_{\pi}(s^n)$$

i.e., the *value function* at state s^n

Value Function: *Recursive Property*

We can further find $p(s^n | s)$ from **transition model** and **policy**

$$\begin{aligned} p_{\pi}(s^n | s) &= \sum_{m=1}^M p(s^n, a^m | s) \\ &= \sum_{m=1}^M p(a^m | s) p(s^n | a^m, s) \\ &= \sum_{m=1}^M \pi(a^m | s) p(s^n | s, a^m) \rightsquigarrow \text{depends on policy} \end{aligned}$$

We know have both terms in terms of **transition model** and **policy**

Value Function: *Recursive Property*

Replacing into the equation, where we left we have

$$\begin{aligned}\mathbb{E}_{\pi} \{G_{t+1} | s\} &= \sum_{n=1}^N \mathbb{E}_{\pi} \{G_{t+1} | s, s^n\} p(s^n | s) \\ &= \sum_{n=1}^N v_{\pi}(s^n) p_{\pi}(s^n | s) \\ &= \sum_{n=1}^N \sum_{m=1}^M v_{\pi}(s^n) p(s^n | s, a^m) \pi(a^m | s)\end{aligned}$$

We can also present it by shorter notation as

$$\mathbb{E}_{\pi} \{G_{t+1} | s\} = \mathbb{E}_{\pi} \{v_{\pi}(S_{t+1}) | s\}$$

Value Function: *Recursive Property*

Back to computation of **value function**, we have

$$\begin{aligned}v_{\pi}(s) &= \bar{\mathcal{R}}_{\pi}(s) + \gamma \mathbb{E}_{\pi} \{G_{t+1} | s\} \\&= \bar{\mathcal{R}}_{\pi}(s) + \gamma \mathbb{E}_{\pi} \{v_{\pi}(S_{t+1}) | s\} \\&= \bar{\mathcal{R}}_{\pi}(s) + \gamma \sum_{n=1}^N v_{\pi}(s^n) p_{\pi}(s^n | s)\end{aligned}$$

This is a **recursive equation** that relates value of one state to other values

*which is a **Bellman equation***

Bellman Equation: Value

Bellman Equation for Value Function

For any policy π the value function at each **state** s satisfies

$$v_{\pi}(s) = \bar{\mathcal{R}}_{\pi}(s) + \gamma \sum_{n=1}^N v_{\pi}(s^n) p_{\pi}(s^n | s)$$

- + Well! What is the **use** of **Bellman equation**?
- It describes a **fixed-point** equation that can be solved for $v_{\pi}(s)$!

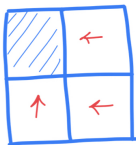
Bellman Equation: *Breaking Down*

$$v_{\pi}(s) = \bar{\mathcal{R}}_{\pi}(s) + \gamma \sum_{n=1}^N v_{\pi}(s^n) p_{\pi}(s^n | s)$$

In general, we have N possible state $\rightsquigarrow$ we have N possible values

- Bellman equation relates each value to other $N - 1$ values
 - ↳ For each s , Bellman equation has N unknowns $v_{\pi}(s^1), \dots, v_{\pi}(s^N)$
- We can write the Bellman equation for all N states
 - ↳ We have N equations each with N unknowns
- We solve this system of equations for unknowns $v_{\pi}(s^1), \dots, v_{\pi}(s^N)$

Example: *Dummy Grid World*



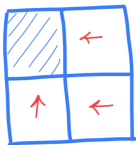
Let's try with our dummy grid world: we saw that

$$\bar{\mathcal{R}}_{\pi}(0) = 0 \quad \bar{\mathcal{R}}_{\pi}(1) = -1 \quad \bar{\mathcal{R}}_{\pi}(2) = -1 \quad \bar{\mathcal{R}}_{\pi}(3) = -1$$

Now let's consider the values *unknown*

$$v_{\pi}(0), v_{\pi}(1), v_{\pi}(2), v_{\pi}(3)$$

Example: *Dummy Grid World*



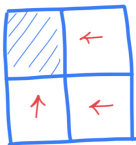
We set $\gamma = 1$ and start with state $s = 0$

$$v_{\pi}(0) = \bar{\mathcal{R}}_{\pi}(0) + \sum_{\bar{s}=0}^3 v_{\pi}(\bar{s}) p_{\pi}(\bar{s}|0)$$

We know that

$$p_{\pi}(\bar{s}|0) = \begin{cases} 1 & \bar{s} = 0 \\ 0 & \bar{s} \neq 0 \end{cases}$$

Example: *Dummy Grid World*

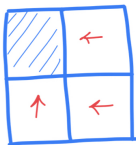


This concludes that at state $s = 0$, *Bellman equation reads*

$$v_{\pi}(0) = 0 + v_{\pi}(0)$$

which is an obvious equation; let's try $s = 1$

Example: *Dummy Grid World*



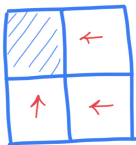
At state $s = 0$, we have

$$v_{\pi}(1) = \bar{\mathcal{R}}_{\pi}(1) + \sum_{\bar{s}=0}^3 v_{\pi}(\bar{s}) p_{\pi}(\bar{s}|1)$$

Again we can easily say based on the *policy* that

$$p_{\pi}(\bar{s}|1) = \begin{cases} 1 & \bar{s} = 0 \\ 0 & \bar{s} \neq 0 \end{cases}$$

Example: Dummy Grid World



This concludes that at state $s = 1$, *Bellman equation reads*

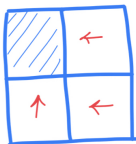
$$v_{\pi}(1) = -1 + v_{\pi}(0)$$

which relates $v_{\pi}(1)$ to $v_{\pi}(0)$. If we keep repeating we get further

$$v_{\pi}(2) = -1 + v_{\pi}(0)$$

$$v_{\pi}(3) = -1 + v_{\pi}(2)$$

Example: *Dummy Grid World*



We now have the system of equations

$$v_{\pi}(1) = -1 + v_{\pi}(0)$$

$$v_{\pi}(2) = -1 + v_{\pi}(0)$$

$$v_{\pi}(3) = -1 + v_{\pi}(2)$$

We also know that $s = 0$ is a **terminal state**, and thus $v_{\pi}(0) = 0$: so, we get

$$v_{\pi}(1) = -1 \quad v_{\pi}(2) = -1 \quad v_{\pi}(3) = -2$$

Bellman Equation: Action-Value

We can find a **Bellman equation** for action-value function as well

$$\begin{aligned} q_{\pi}(s, a) &= \mathbb{E}_{\pi} \{G_t | s, a\} \\ &= \mathbb{E}_{\pi} \{R_{t+1} + \gamma G_{t+1} | s, a\} \\ &= \mathbb{E} \{R_{t+1} | s, a\} + \gamma \mathbb{E}_{\pi} \{G_{t+1} | s, a\} \\ &= \bar{\mathcal{R}}(s, a) + \gamma \underbrace{\mathbb{E}_{\pi} \{G_{t+1} | s, a\}}_{?} \end{aligned}$$

We need to compute

$$\mathbb{E}_{\pi} \{G_{t+1} | s, a\}$$

in terms of the *rewarding-transition model* and *policy*

Action-Value: Recursive Property

We apply the marginalization trick

$$\mathbb{E}_{\pi} \{G_{t+1} | s, a\} = \sum_{n=1}^N \mathbb{E}_{\pi} \{G_{t+1} | S_t = s, S_{t+1} = s^n, A_t = a\} p(s^n | s, a)$$

Attention

Recalling the trajectory of the MDP, we should note that

$$q_{\pi}(s^n, a) \neq \mathbb{E}_{\pi} \{G_{t+1} | S_t = s, S_{t+1} = s^n, A_t = a\} = v_{\pi}(s^n)$$

In fact, once we know S_{t+1} , the **previous action** does not contain any extra information! We only gain information, if we observe A_{t+1} , i.e.,

$$\mathbb{E}_{\pi} \{G_{t+1} | S_t = s, S_{t+1} = s^n, A_{t+1} = a\} = q_{\pi}(s^n, a)$$

Action-Value: *Recursive Property*

So, we can replace it into original equation to get

$$\begin{aligned}\mathbb{E}_{\pi} \{G_{t+1} | s, a\} &= \sum_{n=1}^N \mathbb{E}_{\pi} \{G_{t+1} | S_t = s, S_{t+1} = s^n, A_t = a\} p(s^n | s, a) \\ &= \sum_{n=1}^N v_{\pi}(s^n) p(s^n | s, a)\end{aligned}$$

This implies that

$$\begin{aligned}q_{\pi}(s, a) &= \bar{\mathcal{R}}(s, a) + \gamma \sum_{n=1}^N v_{\pi}(s^n) p(s^n | s, a) \\ &= \bar{\mathcal{R}}(s, a) + \gamma \mathbb{E} \{v_{\pi}(S_{t+1}) | s, a\}\end{aligned}$$

Bellman Equation: Action-Value

Bellman Equation I for Action-Value Function

For any policy π the action-value function at each pair (s, a) satisfies

$$q_{\pi}(s, a) = \bar{\mathcal{R}}(s, a) + \gamma \mathbb{E}_{\pi} \{v_{\pi}(S_{t+1}) | s, a\}$$

Bellman Equation II for Action-Value Function

For any policy π the action-value function at each pair (s, a) satisfies¹

$$q_{\pi}(s, a) = \bar{\mathcal{R}}(s, a) + \gamma \sum_{n=1}^N \sum_{m=1}^M q_{\pi}(s^n, a^m) \pi(a^m | s^n) p(s^n | s, a)$$

¹Check Assignment 1

Computing Action-Value via Bellman Equation

We can again use the recursive equation

$$q_{\pi}(s, a) = \bar{\mathcal{R}}(s, a) + \gamma \sum_{n=1}^N \sum_{m=1}^M q_{\pi}(s^n, a^m) \pi(a^m | s^n) p(s^n | s, a)$$

to find the **action-value function**: we have in this case NM possible values

- **Bellman equation** relates **each action-value** to other **action-values**
 ↳ For **each s and a** , Bellman equation has NM unknowns $q_{\pi}(s^n, a^m)$
- We can write the **Bellman equation** for all NM cases
- We solve this system of equations for **unknowns** $q_{\pi}(s^n, a^m)$